MANOHAR SAI RAMANNA CHOWDARY UPPULURI

Dayton, OH | (937) 604-4651 | Manoharsairamannachowdary@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineer with 2+ years of experience developing and validating enterprise software using C#, C++, Python, and Java. Experienced in object-oriented programming, multithreading, test automation, CI/CD pipelines, and hardware-software integration. Skilled at debugging complex production issues and delivering in Agile teams, with hands-on full-stack (HTML, CSS, JavaScript, Flask) and cloud-native (Docker, Kubernetes, AWS) project experience.

TECHNICAL SKILLS

Languages: C#, C++, Python, Java, JavaScript, SQL
Web & Frameworks: .NET Core, ASP.NET, Flask, Vue.js, HTML, CSS, REST APIs
Cloud & DevOps: AWS (EC2), Docker, Kubernetes, Linux (Ubuntu), Git, GitHub Actions, Azure DevOps, CI/CD
Databases: MySQL, Microsoft SQL Server
Tools: Visual Studio, Jira, Postman
Practices: OOP, Multithreading, Automated Testing, Debugging, Root Cause Analysis, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer | HCL Tech, Andhra Pradesh, India *Dec 2023 – Aug 2024*
Semiconductor Equipment Development Project – Client: Teradyne Inc.
- Developed and verified semiconductor test system software on Teradyne EAGLE platforms using C# .NET, validating hardware-software interactions across multiple deployment cycles.
- Built automated test suites covering 100+ functional, integration, and system scenarios, raising test efficiency 30%.
- Resolved 20+ software and hardware integration defects through log analysis, root cause analysis, and regression tests.
- Owned assigned system bring-up, verification, and first article validation tasks, confirming compliance with hardware and software requirements before production release.
- Partnered with firmware and hardware engineers to validate integrated system functionality and fix cross-layer issues.

Software Engineer | HCL Tech, Chennai, India *Aug 2022 – Nov 2023*
Semiconductor Equipment Sustenance Project – Client: Teradyne Inc.
- Maintained and enhanced test system software across multiple production deployments of integrated systems.
- Automated validation workflows, increasing test coverage and execution consistency across testing cycles.
- Analyzed system logs to resolve integration defects, improving production reliability and reducing downtime.
- Tracked defects and delivered enhancements within sprint timelines in an Agile/Scrum environment using Jira.

Graduate Software Engineer Trainee | HCL Tech, Chennai, India *Jan 2022 – Jul 2022*
- Built C++ and C# .NET skills in object-oriented development for hardware-integrated systems through structured training.
- Assisted senior engineers with validation, debugging, system bring-up, and equipment testing of integrated components.
- Documented technical processes and tracked tasks in Jira, improving team coordination and traceability.

PROJECTS

AI Coding Learning Platform (Code Pathway) | Python, Flask, JavaScript, HTML, CSS, MySQL, AWS EC2 | GitHub
- Built a full-stack platform (JavaScript/HTML/CSS front end, Flask back end) delivering real-time AI code feedback.
- Designed a MySQL database supporting 100+ concurrent users, managing submissions and feedback history.
- Deployed on AWS EC2 with automated monitoring, achieving 99.2% uptime and under 500 ms response time.

Automated Test Framework for Simulated Semiconductor Equipment | C# .NET, C++, GitHub Actions, Azure DevOps
- Developed a test framework executing 100+ test cases per run across 6 hardware protocol simulators via CI/CD.
- Implemented multithreaded test execution, reducing runtime by 31% (about 2.5 hours per cycle).
- Built logging, result tracking, and fault-condition simulations to speed defect analysis and improve coverage.

Kubernetes Microservices Deployment (Ongoing) | Kubernetes, Docker, Minikube, Linux (Ubuntu)
- Containerizing apps with Docker and deploying to Kubernetes via YAML (Deployments, Services, ConfigMaps, Secrets).
- Implementing rolling updates, horizontal scaling, and self-healing deployments; troubleshooting with kubectl.

EDUCATION

Master of Science in Computer Science | Wright State University, OH | GPA: 3.40 *Sep 2024 – May 2026*
B.E. Electronics & Communication | Sathyabama Institute of Science & Technology | GPA: 3.60 *Aug 2018 – Jun 2022*

CERTIFICATIONS & AFFILIATIONS

Certification: AWS Certified Cloud Practitioner (In Progress)
Member: IEEE Computer Society, ACM, Engineers Without Borders (Wright State University)